Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Free-body diagram

Meaning in my words How would you explain free-body diagram to a friend?	**How can you tell?** You are drawing a lamp hanging from a cable. What must be true of every arrow?
Example or sketch What arrows would you draw for a lamp hanging still?	**Non-example and why** A student draws the lamp's downward pull on the cable on the lamp diagram. Where does it belong?

Selected object

Meaning in my words How would you explain selected object to a friend?	**How can you tell?** You circle only the cart. Which force belongs: the hand on the cart or the cart on the hand?
Example or sketch What would you put at the centre of a free-body diagram for a hanging lamp?	**Non-example and why** A student switches to forces on the ceiling halfway through. How should they organize those?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Interacting object

Meaning in my words How would you explain interacting object to a friend?	**How can you tell?** For a book on a table, what objects account for the upward and downward forces?
Example or sketch Which interacting object would you name beside the tension arrow on a hanging lamp?	**Non-example and why** Someone labels a force arrow “the book’s speed.” What interacting object does that name?

Force source

Meaning in my words How would you explain force source to a friend?	**How can you tell?** A floor pushes up on a crate. Which object is the source?
Example or sketch What full label would you write next to the weight arrow on the crate?	**Non-example and why** A student says “right” is the source of a force. What does that word tell us instead?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a boundary around one object. Add force arrows with source labels; put motion information outside the diagram.

Chosen object: _____

My marks mean

Build the force overlay

15

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Keep only forces on the chosen object in its diagram.

CHOOSE ONE JOB TOGETHER

SYSTEMS ENGINEER ☐

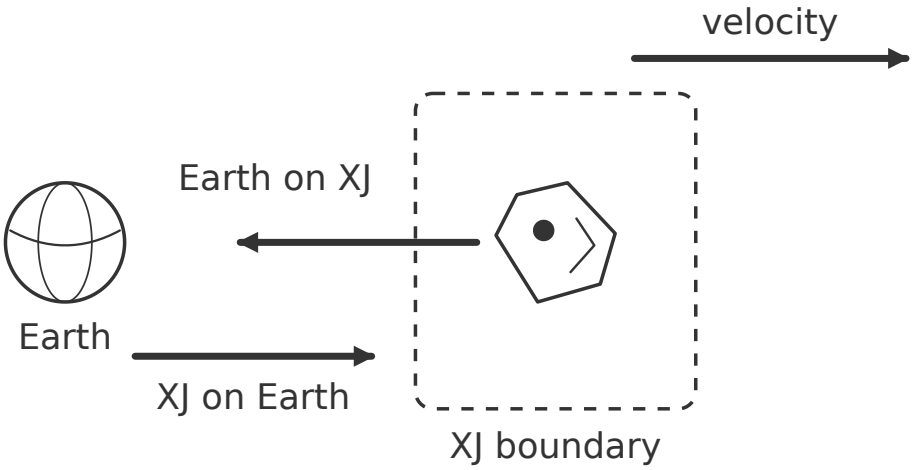
Outline XJ as the chosen object. Keep only the force arrows you would use to calculate XJ's acceleration.

ANIMATION DIRECTOR ☐

Separate the force and motion layers. Tell the animator which Earth arrow belongs in a different shot.

Point to a clue. Keep your first marks; add the update using the color key.

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.



01 OUR FIRST CALL

Which arrows belong in 'forces on XJ'? Explain what you keep and move.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Where does the new arrow belong? Update the force layer and explain.

Switch speakers. Does every arrow act on the object inside your boundary?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks